國立東華大學應用數學系

碩士論文

Exploration and Extension in Spatial Extremes Incorporating Thin-plate Splines Basis Functions

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中華民國 115 年 6 月

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摘要

本論文的核心推論目標，是在未設置監測站的空間位置估計高門檻超標機率。Morris et al. (2017) 爲此提出一套以時空 skew-t 過程爲基礎的階層式貝氏架構，其組成涵蓋四項要素：具 Matérn 協方差結構的 skew-t 邊際分布、可消除遠距漸近相依的每日空間隨機分割、上尾門檻檢查，以及作用於 PIT 變換後時變潛在參數之 AR(1) 先驗。該架構雖以美國環保署 (US EPA) 地面臭氧 75 ppb 合規門檻之預測爲原始應用情境而生，其四要素結構同樣適用於任何重尾、不對稱且具空間相依、推論目標爲上尾超標機率的環境變數。本論文以該架構爲探索對象，追問在此四項要素中「何處」仍保留改善上尾預測精度的空間，並以兩個方向相反的延伸作爲對照探針：時間軸上，將 AR(1) 先驗推廣爲符合 Yule–Walker 關係的穩態 AR(2) 先驗，以保持單位邊際變異數並維繫每日 skew-t 邊際結構；空間軸上，於迴歸成分中引入多重解析度薄板樣條 (MRTS) 基底函數，補捉穩態 Matérn 協方差所無法表現的非定態均值結構。兩項探針於同一 Morris 基準下，以樣本外 Brier 分數與分位數分數於模擬資料、EPA 地面臭氧資料、以及 MERRA-2 地表溫度基準上進行比較。

兩軸結果呈現具方法論意義的不對稱性。MRTS 延伸於若干上尾分位水準下確實降低 Brier 分數；AR(2) 延伸於 Brier 分數上則未能系統性優於 AR(1)。AR(2) 的負向發現反而是更具方法學價值的結果：本文於 Discussion 中以三條命題證明，爲維繫邊際 skew-t 結構所必要的單位變異數標準化，同時亦使所有潛在過程在時間 t 的邊際對自回歸係數 ϕ 不變；因此樣本外 Brier 積分在主要途徑上爲 ϕ 盲，殘餘的後驗時間借力效益受到一個 ϕ 不變的空間雜訊下界所限制。此論證不僅適用於 AR(2)，亦適用於任何在 PIT 邊際保留條件下的標準化 AR(p) 延伸。本論文之可移植結論爲一項方向性的方法學指引，其適用範圍超出臭氧此一特定應用：於 Morris 核與其保留標準化的變體之下，樣本外上尾 Brier 預測之提升空間實際位於空間迴歸成分，將方法學努力用於擴充時間先驗在結構上受到限制。

關鍵字：時空 skew-t 過程、門檻超越預測、空間隨機分割、階層式貝氏推論、標準化 AR(2) 先驗、多重解析度薄板樣條、Brier 分數、臭氧

Abstract

Spatial prediction of high-threshold exceedances at unmonitored locations is the central inferential target of this thesis. Morris et al. (2017) developed a hierarchical Bayesian spatio-temporal skew-t framework for this task, combining four ingredients: a skew-t marginal with stationary Matérn covariance, a daily random spatial partition that removes long-range asymptotic dependence, upper-tail censoring, and an AR(1) prior on PIT-transformed time-varying latents. The framework was demonstrated on U.S. EPA ground-level ozone above the 75-ppb compliance threshold, but its four-ingredient structure carries over to any heavy-tailed, asymmetric, spatially-correlated environmental field whose policy-relevant target is an upper-tail exceedance probability. This thesis is an exploratory study of that framework. We ask where, among its four ingredients, the remaining room for improving upper-tail prediction lives, and probe two extension axes that pull in opposite directions: on the temporal axis, a standardized AR(2) prior on the time-varying latents that retains the unit marginal variance and hence the skew-t margin at every day via Yule–Walker constraints; on the spatial axis, multi-resolution thin-plate spline (MRTS) basis functions added to the regression component to capture non-stationary mean structure that lies beyond the reach of a stationary covariance. Both probes are evaluated against the same Morris baseline on simulated data, on EPA ground-level ozone observations, and on a controlled MERRA-2 surface-temperature benchmark, using held-out Brier and quantile scores.

The contrast yields a directional answer with a methodological asymmetry. The MRTS extension reduces the upper-tail Brier score at selected exceedance levels; the AR(2) extension does not systematically improve over AR(1). The negative AR(2) finding is the more informative result: a chain of three propositions in the Discussion shows that the unit-variance standardization required to preserve the marginal skew-t structure also renders the time-t latent marginal invariant to the autoregressive coefficients ϕ, that the held-out Brier integral sees the latent field only through this ϕ-blind marginal, and that the residual posterior-pooling benefit is bounded below by a ϕ-invariant spatial-noise floor. The argument applies to any standardized AR(p) extension of a hierarchical extreme-value model with PIT-preserved margins. The transferable conclusion is a piece of directional methodological guidance for upper-tail spatial prediction beyond ozone alone: in this class of models, the leverage on held-out Brier prediction lives in the spatial regression component, and methodological effort spent enriching the temporal prior is structurally bounded in its payoff.

Keywords: spatio-temporal skew-t process; threshold exceedance prediction; random spatial partition; hierarchical Bayesian inference; standardized AR(2) prior; multi-resolution

thin-plate splines; Brier score; ozone

Contents

1 Introduction

Across environmental statistics, the policy-relevant inferential target is often not the average behavior of a field but the probability that it exceeds a high threshold at unmonitored locations. Ground-level ozone is the canonical example: under the U.S. Environmental Protection Agency National Ambient Air Quality Standards, a monitoring site is in compliance if its three-year average of the annual fourth-highest daily maximum eight-hour concentration does not exceed 75 parts per billion (ppb), so the inferential target is a spatial probability statement about the upper tail of a daily concentration field. The same target arises whenever a heavy-tailed, asymmetric environmental variable is regulated by an upper threshold—daily precipitation extremes for flood-risk planning, surface temperature for heat-warning levels, and similar applications elsewhere. Monitoring networks in these settings are typically sparse and unevenly distributed in space, so the prediction problem cannot be answered from raw station data alone; it requires a spatial model whose marginal and dependence behavior is faithful to the upper tail of the observed field.

Off-the-shelf Gaussian spatial models are unsuitable for this task on three counts. First, fitting a Gaussian process to the entire record forces low and moderate observations to dictate the covariance fit, so the implied predictive distribution at the upper tail is badly biased. Second, the Gaussian distribution is light-tailed and symmetric, whereas heavy-tailed environmental fields are typically right-skewed. Third, two locations $\mathbf{s}_1, \mathbf{s}_2$ drawn from a Gaussian process satisfy $\lim_{c\to\infty} \Pr[Y(\mathbf{s}_2) > c \mid Y(\mathbf{s}_1) > c] = 0$ whenever $|\mathrm{Cor}(Y(\mathbf{s}_1), Y(\mathbf{s}_2))| < 1$—the Gaussian process is asymptotically independent at every positive distance—yet real environmental fields routinely produce synoptic-scale extreme days on which a large geographic region sits above its 90th percentile simultaneously. A model that drives joint exceedance probabilities to zero at high thresholds is structurally incapable of representing that signal, so a different distributional family is needed.

The classical alternative is to discard the bulk of the record and apply extreme-value theory to the exceedances or block maxima that remain, using the generalized Pareto and generalized extreme-value distributions and their spatial generalizations through max-stable processes. Two drawbacks make that route ill-suited to the present setting. Full-likelihood inference for max-stable processes is intractable beyond a handful of sites, and the techniques developed to circumvent this—pairwise composite likelihoods and recent full-likelihood approximations—do not scale gracefully to monitoring networks of hundreds or thousands of sites. More fundamentally, max-stable models are valid only above a sufficiently high threshold; for many regulatory targets the policy-relevant threshold sits near the bulk of the daily distribution rather than well into the tail (at the 75 ppb ozone level, for example, the threshold lies below the 90th percentile at hundreds of EPA monitors), so the asymptotic regime that justifies max-stable modeling is not in effect. The methodological gap is therefore the middle ground: a model that respects heavy tails and asymptotic dependence without committing to the asymptotic regime that pure extreme-value theory presumes.

Morris et al. (2017) proposed a hierarchical Bayesian framework designed for exactly this middle ground. The model has four ingredients. (i) A spatial skew-t process replaces the Gaussian assumption, delivering asymmetric, heavy-tailed marginals and—unlike the Gaussian—nontrivial asymptotic dependence between nearby sites. (ii) A stationary isotropic Matérn covariance with nugget controls the spatial smoothness of the underlying Gaussian factor. (iii) A daily random spatial partition—induced by latent knots and a Voronoi assignment—lets the scale and skew latents vary across the domain, breaking the unphysical long-range asymptotic dependence that a single shared skew and scale would otherwise impose. (iv) Upper-tail censoring fits the model only to observations above a chosen threshold T, so low and moderate days no longer distort the tail fit. The four time-varying latent parameters per knot are made temporally coherent by a stationary AR(1) prior on each PIT-transformed copy, preserving the marginal skew-t structure at every day while still borrowing strength across days, and the framework is fitted by Markov chain Monte Carlo at scales of hundreds to thousands of sites. Although Morris demonstrated the framework on ozone, its four-ingredient structure is generic and transfers directly to any heavy-tailed, asymmetric, spatially-correlated environmental field whose policy target is an upper-tail exceedance probability.

This thesis treats the Morris framework as the prototype for that broader class of models and asks where in its four-ingredient structure the remaining room for improving upper-tail prediction actually lives. Two extension axes pull in opposite directions on the hierarchy: on the temporal axis, the AR(1) prior on each PIT-transformed latent is generalized to a standardized AR(2) prior whose Yule–Walker normalization preserves the unit marginal variance and hence the spatial skew-t margin at every day; on the spatial axis, the regression component is augmented with multi-resolution thin-plate spline basis functions (Tzeng and Huang, 2018) that capture non-stationary mean structure beyond the reach of a stationary covariance. Both probes are evaluated against the same Morris baseline on simulated data, on EPA ground-level ozone, and on a controlled MERRA-2 surface-temperature benchmark, using held-out Brier and quantile scores. The contrast yields a directional answer with a methodological asymmetry: the MRTS extension reduces the upper-tail Brier score at selected exceedance levels, while the AR(2) extension does not systematically improve over AR(1). The negative

AR(2) finding is the more methodologically informative result; the Discussion traces it to a structural feature of the Morris kernel that generalizes to any standardized AR(p) extension of a hierarchical extreme-value model with PIT-preserved margins, and the transferable conclusion is that the leverage on held-out upper-tail prediction in this class of models lives in the spatial regression component rather than in the temporal prior.

The remainder of the thesis is organized as follows. The Literature review covers the two pieces of prior work the thesis builds on: Section **??** summarizes the Morris baseline and Section **??** introduces the MRTS basis. The Methodology states the two extensions in a common notation (Section **??**, Section **??**) and defines the Brier and quantile scoring rules used throughout (Section **??**). The Simulation study reports two design–results pairs: Section **??**–Section **??** compare the MRTS variant against the Morris baseline on the Morris benchmark, and Section **??**–Section **??** compare the AR(2) variant against the AR(1) baseline on data with known AR(2) dependence. The Data analysis section then validates both probes on real data: Section **??** runs the full candidate model class on EPA ground-level ozone, and Section **??** runs the temporal probe on a MERRA-2 surface-temperature benchmark. The thesis closes with a Discussion that gives the mathematical explanation, structured around three propositions, for why the AR(2) extension cannot systematically improve the held-out Brier score in this class of models.

2 Literature review

2.1 The spatio-temporal skew-t model

The thesis treats the spatio-temporal skew-t model of Morris et al. (2017) as the object of investigation. We summarize that baseline here so that the two probe extensions developed in the Methodology section—one on the temporal axis (AR(2)) and one on the spatial-mean axis (MRTS)—can be stated against a common notation.

Let $Y_t(\mathbf{s})$ denote the response at spatial location $\mathbf{s} \in \mathcal{D} \subset \mathcal{R}^2$ on day $t \in \{1, \ldots, n_t\}$. Following Morris et al. (2017), we model

$$Y_t(\mathbf{s}) \;=\; \mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta} \;+\; \lambda\, \sigma_t(\mathbf{s})\, |z_t(\mathbf{s})| \;+\; \sigma_t(\mathbf{s})\, v_t(\mathbf{s}), \tag{1}$$

where $\mathbf{X}_t(\mathbf{s})$ is a p-vector of covariates, $\boldsymbol{\beta}$ is the corresponding regression coefficient, $\lambda \in \mathcal{R}$ is a scalar skewness parameter, and $v_t(\mathbf{s})$ is a standard Gaussian process with stationary isotropic

Matérn correlation

$$\rho(h) \;=\; (1-\gamma)\, I(h=0) \;+\; \gamma \,\frac{1}{\Gamma(\nu)\, 2^{\nu-1}} \left(\sqrt{2\nu}\,\frac{h}{\varphi}\right)^{\nu} K_\nu\!\left(\sqrt{2\nu}\,\frac{h}{\varphi}\right), \tag{2}$$

where $\varphi > 0$ is the spatial range, $\nu > 0$ is the smoothness, $\gamma \in [0,1]$ is the proportion of variance attributed to the spatial component, K_ν is the modified Bessel function of the second kind, and $h = \|\mathbf{s}_1 - \mathbf{s}_2\|$. Marginalizing over $|z_t(\mathbf{s})|$ and $\sigma_t(\mathbf{s})$, the finite-dimensional vector $\mathbf{Y}_t = [Y_t(\mathbf{s}_1), \ldots, Y_t(\mathbf{s}_n)]^\top$ follows an n-dimensional skew-t distribution; consequently (**??**) provides asymmetric and heavy-tailed margins together with asymptotic dependence between nearby sites.

To remove the long-range asymptotic dependence that arises when a single scalar z and σ^2 are shared across the entire domain, the model uses a daily random partition. For each day t, let $\mathbf{w}_{t1}, \ldots, \mathbf{w}_{tK}$ denote K spatial knots in $\mathcal{D}$, and define the partition subregions

$$P_{tk} \;=\; \{\mathbf{s} \in \mathcal{D} : k = \arg\min_\ell \|\mathbf{s} - \mathbf{w}_{t\ell}\|\}, \qquad k = 1, \ldots, K. \tag{3}$$

For $\mathbf{s} \in P_{tk}$ we set $z_t(\mathbf{s}) = z_{tk}$ and $\sigma_t^2(\mathbf{s}) = \sigma_{tk}^2$, with $z_{tk} \overset{iid}{\sim} N(0,1)$ and $\sigma_{tk}^2 \overset{iid}{\sim} \mathrm{IG}(a/2, b/2)$. Within each subregion the model retains the skew-t form, while sites in different partitions are not asymptotically dependent.

Because the EPA compliance criterion focuses on exceedances of a high level L, the model fits only the upper tail by censoring the data at a pre-specified threshold $T \le L$. Writing the partially censored response $\widetilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$, inference is based on $\widetilde{\mathbf{Y}}_t = [\widetilde{Y}_t(\mathbf{s}_1), \ldots, \widetilde{Y}_t(\mathbf{s}_n)]^\top$, and the censored values below T are imputed inside the MCMC.

Daily temporal dependence is introduced on the latent parameters that drive the tail, namely the knot locations $\mathbf{w}_{tk}$, the skewness latents z_{tk}, and the scale parameters σ_{tk}^2. To preserve the marginal skew-t distribution at every day, each parameter is first transformed to a standard normal scale via the probability integral transform:

$$w_{tki}^* = \Phi^{-1}\left[\frac{w_{tki} - \min(\mathbf{s}_i)}{\max(\mathbf{s}_i) - \min(\mathbf{s}_i)}\right], \quad i = 1, 2, \tag{4}$$

$$z_{tk}^* = \Phi^{-1}\{\mathrm{HN}(z_{tk})\}, \tag{5}$$

$$\sigma_{tk}^{2*} = \Phi^{-1}\{\mathrm{IG}(\sigma_{tk}^2)\}, \tag{6}$$

where Φ is the standard normal distribution function, HN is the half-normal distribution function, and IG is the inverse-gamma distribution function with parameters $(a/2, b/2)$. After transformation, each $\mathbf{w}_{tk}^*, z_{tk}^*, \sigma_{tk}^{2*}$ has a standard normal marginal distribution, and Morris et al. (2017) place a stationary AR(1) Gaussian time series on each of them:

$$\mathbf{w}_{tk}^* \mid \mathbf{w}_{t-1,k}^* \sim N_2\big(\phi_w \, \mathbf{w}_{t-1,k}^*,\ (1-\phi_w^2)\,\mathbf{I}_2\big)\,, \tag{7}$$
$$z_{tk}^* \mid z_{t-1,k}^* \sim N\big(\phi_z \, z_{t-1,k}^*,\ 1-\phi_z^2\big)\,, \tag{8}$$
$$\sigma_{tk}^{2*} \mid \sigma_{t-1,k}^{2*} \sim N\big(\phi_\sigma \, \sigma_{t-1,k}^{2*},\ 1-\phi_\sigma^2\big)\,, \tag{9}$$

with $|\phi_w|, |\phi_z|, |\phi_\sigma| < 1$, and initial values $\mathbf{w}_{1k}^* \sim N_2(\mathbf{0}, \mathbf{I}_2)$, $z_{1k}^* \sim N(0,1)$, $\sigma_{1k}^{2*} \sim N(0,1)$. Transforming back through (**??**)–(**??**) gives $\mathbf{w}_{tk} \sim \mathrm{Unif}(\mathcal{D})$, $z_{tk} \sim \mathrm{HN}(0,1)$, and $\sigma_{tk}^2 \sim \mathrm{IG}(a/2, b/2)$ at every t, so that the spatial skew-t structure of (**??**) is preserved at each day. The complete hierarchical model is fitted via MCMC with Gibbs and Metropolis–Hastings steps, with predictions at unobserved sites obtained by Bayesian kriging on the multivariate normal full conditional of $\mathbf{Y}_t$.

This skew-t baseline—partitioned, censored, and AR(1)-correlated in time—defines the reference specification against which the two exploratory probes of this thesis are evaluated. The temporal probe extends the AR(1) prior on the PIT-transformed latents and is developed methodologically in Section **??**; the spatial probe rests on the multi-resolution thin-plate spline basis introduced in the next subsection.

2.2 Multi-resolution thin-plate splines basis functions

Multi-resolution thin-plate spline (MRTS) basis functions provide a smoothness-ordered basis on a two-dimensional spatial domain whose first three components reproduce the constant and linear spatial trend and whose remaining components encode progressively finer-scale spatial structure. The thesis uses this basis to enrich the regression component of the spatial skew-t model with non-stationary mean structure that the stationary Matérn correlation (**??**) cannot represent.

According to Tzeng and Huang (2018), the multi-resolution spline basis function $f_k(\mathbf{s})$ is defined as follows. Fix d control points $\mathbf{c}_i = (x_{i1}, x_{i2})^\top \in \mathcal{R}^2$ for $i = 1, \ldots, d$, and let

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} \\ \vdots & \vdots & \vdots \\ 1 & x_{d1} & x_{d2} \end{pmatrix} \tag{10}$$

be the $d \times 3$ design matrix at the control points. Then

$$f_k(\mathbf{s}) \; = \; \begin{cases} 1, & k = 1, \\ x_{k-1}, & k = 2, 3, \\ \lambda_{k-3}^{-1} \big\{ \boldsymbol{\phi}(\mathbf{s}) \; - \; \boldsymbol{\Phi}\, \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}\, \mathbf{x} \big\}^\top \mathbf{v}_{k-3}, & k = 4, \ldots, d, \end{cases} \tag{11}$$

where $\mathbf{x} = (1, \mathbf{s}^\top)^\top = (1, x_1, x_2)^\top$, $\boldsymbol{\phi}(\mathbf{s}) = (\phi_1(\mathbf{s}), \ldots, \phi_d(\mathbf{s}))^\top$ with

$$\phi_i(\mathbf{s}) \; = \; \frac{1}{8\pi} \, \|\mathbf{s} - \mathbf{c}_i\|^2 \, \log\big(\|\mathbf{s} - \mathbf{c}_i\|\big), \qquad i = 1, \ldots, d, \tag{12}$$

$\boldsymbol{\Phi}$ is the $d \times d$ matrix with (i, j)-th element $\phi_j(\mathbf{c}_i)$, $\mathbf{v}_k$ is the k-th column of $\mathbf{V}$, and $\mathbf{V}\,\mathrm{diag}(\lambda_1, \ldots, \lambda_d)\,\mathbf{V}^\top$ is the eigendecomposition of $\mathbf{Q}\,\boldsymbol{\Phi}\,\mathbf{Q}$ with $\lambda_1 \geq \cdots \geq \lambda_d$ and $\mathbf{Q} = \mathbf{I} - \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$. In this context $\mathbf{c}_1, \ldots, \mathbf{c}_d$ are referred to as control points. The selection of control points influences the smoothness and complexity of the resulting spline functions, and directly determines their local versus global behavior: placing control points densely in regions of interest enhances feature capture in those areas, so the same construction can be used both for globally smooth enrichment and for locally adaptive corrections.

The first three basis functions f_1, f_2, f_3 reproduce the constant and linear spatial trend; the remaining functions $f_4, \ldots, f_d$ are smoothness-ordered through the eigenvalues $\lambda_1 \geq \cdots \geq \lambda_{d-3}$, with f_4 carrying the largest-scale residual spatial structure and f_d the finest. Whenever the baseline regression $\mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta}$ in (**??**) already contains an intercept and a first-order spatial trend—as it does in every experiment of this thesis—only the smooth components $f_4(\mathbf{s}), f_5(\mathbf{s}), \ldots$ contribute new spatial information. The exact augmentation used inside the spatial skew-t model is given in Section **??**.

3 Methodology

3.1 Incorporating MRTS covariates

The MRTS basis of Section **??** enters the Morris hierarchy through the regression component, leaving the latent partition and the AR(1) (or AR(2)) priors on $\mathbf{w}^*_{tk}, z^*_{tk}, \sigma^{2*}_{tk}$ untouched. Because every baseline regression in this thesis already contains an intercept and a first-order spatial trend, the polynomial part of the MRTS basis (f_1, f_2, f_3) is omitted from the augmentation to avoid duplication; only the smooth components $f_4, f_5, \ldots$ contribute new spatial information. Fix the number of additional spline covariates K_{MRTS} and let

$$(\mathbf{s}) \;=\; \big(f_4(\mathbf{s}),\, f_5(\mathbf{s}),\, \ldots,\, f_{K_{\text{MRTS}}+3}(\mathbf{s})\big)^\top \tag{13}$$

collect the next K_{MRTS} smooth basis evaluations defined in (**??**). Replace the day-t covariate vector $\mathbf{X}_t(\mathbf{s})$ in (**??**) with the augmented vector

$$\mathbf{X}_t^*(\mathbf{s}) \;=\; \big[\mathbf{X}_t(\mathbf{s})^\top,\; (\mathbf{s})^\top\big]^\top, \tag{14}$$

and the regression coefficient $\boldsymbol{\beta}$ with the matching augmented vector $\boldsymbol{\beta}^* = (\boldsymbol{\beta}^\top, \boldsymbol{\alpha}^\top)^\top$, where $\boldsymbol{\alpha} \in \mathcal{R}^{K_{\text{MRTS}}}$ holds the MRTS coefficients. The kernel (**??**) then reads

$$Y_t(\mathbf{s}) \;=\; \mathbf{X}_t^*(\mathbf{s})^\top \boldsymbol{\beta}^* \;+\; \lambda\, \sigma_t(\mathbf{s})\, |z_t(\mathbf{s})| \;+\; \sigma_t(\mathbf{s})\, v_t(\mathbf{s}), \tag{15}$$

which is identical to the Morris specification (**??**) on all components other than the mean. Setting $\boldsymbol{\alpha} = \mathbf{0}$ recovers the unenriched Morris regression, so the MRTS extension nests the baseline and adds K_{MRTS} additional regression coefficients. The basis evaluations $f_k(\mathbf{s})$ are precomputed once from (**??**)–(**??**) using a fixed grid of control points and treated as known covariates inside the MCMC; the `autoFRK` R package (Tzeng and Huang, 2018) supplies these evaluations. Because the MRTS extension only changes the mean component, the censoring layer $\widetilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$ and the inverse-PIT transforms (**??**)–(**??**) that drive the latent hierarchy carry over verbatim. The number of additional spline covariates K_{MRTS} is the practical tuning parameter: Section **??** compares the choices $K_{\text{MRTS}} \in \{10, 15, 20\}$ on the Morris benchmark.

3.2 Extension to AR(2) time-varying coefficients

The AR(1) prior in (**??**)–(**??**) is one of the four tuning choices in the Morris hierarchy, and its tightness is precisely what we want to probe along the temporal axis. Replacing it with a stationary AR(2) prior is the minimal extension that keeps the unit-marginal-variance constraint of the inverse PIT (**??**)–(**??**) active while admitting one additional autoregressive lag and the oscillatory persistence inside the AR(2) stationarity triangle that an AR(1) cannot represent. The construction below is designed so that the AR(1) prior is recovered exactly as the $\phi_2 \to 0$ limit, so the two priors can be compared on equal footing.

The AR(2) recursion. For a generic transformed parameter $\{Y_t\}_{t=1}^{n_t}$ – standing in for any of the scalar components of $\mathbf{w}^*_{tk}$, or for z^*_{tk} or σ^{2*}_{tk} at a fixed knot k – the AR(2) prior is

$$Y_t \;=\; \phi_1\, Y_{t-1} \;+\; \phi_2\, Y_{t-2} \;+\; \epsilon_t, \qquad \epsilon_t \overset{iid}{\sim} N(0, \sigma^2_\epsilon), \qquad t \geq 3. \tag{16}$$

For the recursion to define a stationary process, (ϕ_1, ϕ_2) must lie inside the stationarity triangle

$$|\phi_2| < 1, \qquad \phi_1 + \phi_2 < 1, \qquad \phi_2 - \phi_1 < 1. \tag{17}$$

Just as in the AR(1) case, the AR(2) parameters $(\phi_1^{(w)}, \phi_2^{(w)})$, $(\phi_1^{(z)}, \phi_2^{(z)})$, and $(\phi_1^{(\sigma)}, \phi_2^{(\sigma)})$ are separately defined for $\mathbf{w}^*_{tk}$, z^*_{tk}, and σ^{2*}_{tk}.

Yule–Walker moments under unit marginal variance. We require $\gamma_0 := \mathrm{Var}(Y_t) = 1$ so that the inverse PIT applied to Y_t recovers the half-normal, inverse-gamma, or uniform target margins on the original scale. Multiplying (**??**) by Y_{t-h} and taking expectations yields the Yule–Walker recursion

$$\gamma_h \;=\; \phi_1\, \gamma_{h-1} \;+\; \phi_2\, \gamma_{h-2}. \tag{18}$$

Setting $h = 1$ and using the symmetry $\gamma_{-1} = \gamma_1$ gives

$$\gamma_1 \;=\; \frac{\phi_1}{1 - \phi_2}, \qquad \gamma_2 \;=\; \phi_1\, \gamma_1 + \phi_2, \tag{19}$$

and multiplying (**??**) by Y_t pins down the innovation variance

$$\sigma^2_\epsilon \;=\; 1 - \phi_1\, \gamma_1 - \phi_2\, \gamma_2. \tag{20}$$

Setting $\phi_2 = 0$ in (**??**)–(**??**) recovers the AR(1) form $\gamma_1 = \phi_1$, $\sigma^2_\epsilon = 1 - \phi_1^2$, so the AR(2) prior strictly contains the AR(1) prior used by Morris et al. (2017).

Stationary joint of the initial pair. Because the recursion (**??**) requires two seeds, the joint distribution of (Y_1, Y_2) has to be specified separately. Independent $N(0, 1)$ initial values would create a transient regime in which $\mathrm{Var}(Y_t) \neq 1$ for the first several days, breaking the PIT mechanism that produces the correct skew-t margin. Stationarity therefore forces the bivariate Gaussian initial condition

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{pmatrix} \right), \tag{21}$$

which yields the conditional density used when updating Y_2 alone:

$$Y_2 \mid Y_1 \sim N\big(\gamma_1 Y_1,\ 1 - \gamma_1^2\big)\,. \tag{22}$$

Note that the mean of (**??**) is $\gamma_1 Y_1$, not $\phi_1 Y_1$, and its variance is $1 - \gamma_1^2$, not σ_ϵ^2; the two expressions coincide only when $\phi_2 = 0$. In the AR(2) implementation it is essential that both the simulator and the MCMC density at $t = 2$ use (**??**) so that the marginal property $\text{Var}(Y_t) = 1$ holds at every t.

Full conditional with three prior factors. The AR(2) Markov structure factorizes the joint prior of $(Y_1, \ldots, Y_{n_t})$ as

$$p(Y_1, \ldots, Y_{n_t}) = p(Y_1, Y_2) \prod_{s=3}^{n_t} p(Y_s \mid Y_{s-1}, Y_{s-2}), \tag{23}$$

with $p(Y_1, Y_2)$ given by (**??**) and each conditional factor for $s \geq 3$ following (**??**). For an interior time point $3 \leq t \leq n_t - 2$, the value Y_t enters (**??**) through three factors: the self factor $p(Y_t \mid Y_{t-1}, Y_{t-2})$, the lag-1 factor $p(Y_{t+1} \mid Y_t, Y_{t-1})$, and the lag-2 factor $p(Y_{t+2} \mid Y_{t+1}, Y_t)$. The full conditional in the Gibbs-within-MH update of Y_t is therefore

$$\pi(Y_t \mid Y_{-t}, \mathbf{Y}) \propto L(\mathbf{Y} \mid Y_t) \cdot \underbrace{p(Y_t \mid Y_{t-1}, Y_{t-2})}_{\text{self}} \cdot \underbrace{p(Y_{t+1} \mid Y_t, Y_{t-1})}_{\text{lag-1}} \cdot \underbrace{p(Y_{t+2} \mid Y_{t+1}, Y_t)}_{\text{lag-2}}, \tag{24}$$

where $L(\mathbf{Y} \mid Y_t)$ is the spatial likelihood contributed by Y_t through the partition and Gaussian process structure of (**??**). The lag-2 factor is absent for $t = n_t - 1$ and both lag-1 and lag-2 factors are absent for $t = n_t$. Under a symmetric Gaussian random-walk proposal $Y_t^{(c)} \sim N(Y_t, s^2)$, the Metropolis–Hastings acceptance ratio is

$$R = \min\left(1,\ \frac{\pi(Y_t^{(c)} \mid Y_{-t}, \mathbf{Y})}{\pi(Y_t \mid Y_{-t}, \mathbf{Y})}\right), \tag{25}$$

in which all three prior factors must appear; omitting the lag-2 factor distorts the stationary distribution of the chain away from the AR(2) joint and produces a lag-2 autocorrelation that does not match the Yule–Walker value in (**??**).

Application to all three transformed parameters. The AR(2) prior (**??**) together with the stationary initial condition (**??**) replaces (**??**)–(**??**) for each transformed parameter:

$$\mathbf{w}^*_{tk} \mid \mathbf{w}^*_{t-1,k}, \mathbf{w}^*_{t-2,k} \sim N_2\Big(\phi_1^{(w)}\, \mathbf{w}^*_{t-1,k} + \phi_2^{(w)}\, \mathbf{w}^*_{t-2,k},\ \sigma^2_{\epsilon,w}\, \mathbf{I}_2\Big)\,, \tag{26}$$

$$z^*_{tk} \mid z^*_{t-1,k}, z^*_{t-2,k} \sim N\Big(\phi_1^{(z)}\, z^*_{t-1,k} + \phi_2^{(z)}\, z^*_{t-2,k},\ \sigma^2_{\epsilon,z}\Big)\,, \tag{27}$$

$$\sigma^{2*}_{tk} \mid \sigma^{2*}_{t-1,k}, \sigma^{2*}_{t-2,k} \sim N\Big(\phi_1^{(\sigma)}\, \sigma^{2*}_{t-1,k} + \phi_2^{(\sigma)}\, \sigma^{2*}_{t-2,k},\ \sigma^2_{\epsilon,\sigma}\Big)\,, \tag{28}$$

where each pair $(\phi_1^{(\cdot)}, \phi_2^{(\cdot)})$ lies inside the stationarity triangle (**??**), the innovation variances are obtained from (**??**)–(**??**), and the initial pairs $(\mathbf{w}^*_{1k}, \mathbf{w}^*_{2k})$, (z^*_{1k}, z^*_{2k}), $(\sigma^{2*}_{1k}, \sigma^{2*}_{2k})$ follow the bivariate Gaussian (**??**) component-wise. After inverse transformation through (**??**)–(**??**), the back-transformed parameters $\mathbf{w}_{tk}, z_{tk}, \sigma^2_{tk}$ retain the same marginal distributions as in the AR(1) baseline, so that the spatial skew-t model (**??**) is preserved at every t. Setting $\phi_2^{(\cdot)} = 0$ in (**??**)–(**??**) recovers the Morris AR(1) specification, so the AR(2) extension nests the original time-series model and adds one additional autoregressive coefficient per latent process.

3.3 Predictive criterion: the Brier score

Because the inferential target of this thesis is the probability of upper-tail exceedances at unmonitored sites, the primary scoring rule used throughout the simulation study and the data analysis is the Brier score (Gneiting and Raftery, 2007). The Brier score is a strictly proper score for binary events—predicting Pr$(Y > L)$ correctly minimizes the expected score in any direction other than the truth—which makes it the appropriate metric when the goal is the calibrated estimation of exceedance probabilities at policy-relevant thresholds. For an exceedance level L, the Brier score is

$$\mathrm{BS}(L) \;=\; \big\{\, I(y > L) \;-\; \widehat{P}(Y > L)\,\big\}^2, \tag{29}$$

where y is the held-out observation and $\widehat{P}(Y > L)$ is the posterior predictive probability of exceeding L. The score is averaged over all held-out sites and time points, and lower values indicate better exceedance prediction.

For complementary diagnostic purposes the experiments also report the quantile score at level α (Gneiting and Raftery, 2007), which evaluates the accuracy of the posterior predictive α-quantile and is likewise smaller when prediction is better. The reported exceedance and quantile levels are concentrated in the upper tail of the marginal response distribution: $q(0.90)$, $q(0.95)$, $q(0.98)$, $q(0.99)$, and—in selected MRTS comparisons—$q(0.995)$. With the scoring rule fixed, the Simulation study that follows can be read as a controlled experiment that asks which of the two probe extensions actually moves these scores.

4 Simulation study

The simulation study is the empirical half of the exploratory comparison: it is designed to localize whether and where the two probe extensions reduce the upper-tail Brier score relative to the space-time skew-t threshold model of Morris et al. (2017), and—by construction of the data-generating settings—to give each probe its most favorable test. The section reports two experiments in two design–results pairs. Section **??**–Section **??** compare MRTS-augmented analysis models against the Morris baseline on the Morris benchmark, and Section **??**–Section **??** compare AR(2)-augmented analysis models against the AR(1) baseline on data with known AR(2) dependence. Both experiments score predictions with the held-out Brier and quantile scores defined in Section **??**.

4.1 MRTS on the Morris benchmark: design

The benchmark simulation follows the structure of Morris et al. (2017): observations are generated at $n_s = 144$ spatial sites, sampled uniformly on $[0, 10] \times [0, 10]$, over $n_t = 50$ independent time points. The mean structure is constant in the data generation, $\mathbf{X}^\top \boldsymbol{\beta} = 10$, while the fitted models include an intercept and first-order spatial trend. The main data-generating settings include Gaussian data, skew-t data with one knot, skew-t data with five knots, a max-stable setting, and a Brown–Resnick max-stable setting. For each setting, 50 simulated data sets are generated, with 100 sites used for training and 44 sites withheld for validation on each data set.

For the baseline comparison, five Bayesian models are considered:

1. Gaussian marginal distribution with $K = 1$ knot;

2. Skew-t marginal distribution with $K = 1$ knot and no thresholding;

3. Symmetric-t marginal distribution with $K = 1$ knot and threshold $T = q(0.80)$;

4. Skew-t marginal distribution with $K = 5$ knots and no thresholding;

5. Symmetric-t marginal distribution with $K = 5$ knots and threshold $T = q(0.80)$.

Here $q(0.80)$ denotes the empirical 80th percentile of the simulated response. These five models represent the main combinations of tail behavior, thresholding, and spatial partitioning used in the Morris model.

To evaluate the MRTS covariate extension, the five baseline models are each compared against an MRTS-augmented variant in which the regression component is enriched by the basis evaluations $f_1(\mathbf{s}), \ldots, f_{K_{\text{MRTS}}}(\mathbf{s})$ of Section **??**. We focus on simulation settings 4 and 5: setting 4 is a one-knot skew-t data-generating process with $\lambda = 3$, and setting 5 is a five-knot partitioned skew-t data-generating process with $\lambda = 3$. For setting 4, each baseline model is compared with versions using $K_{\text{MRTS}} \in \{10, 15, 20\}$; for setting 5, each baseline model is compared with a version using $K_{\text{MRTS}} = 15$. Each setting uses 50 simulated data sets and the same $100/44$ train/validation split as the baseline benchmark, with each MCMC chain run for $20\,000$ iterations with $10\,000$ burn-in iterations.

4.2 MRTS on the Morris benchmark: results

The benchmark simulation supports the main conclusion of Morris et al. (2017): when the fitted model matches the data-generating mechanism, it tends to improve predictive performance, and when the data are non-Gaussian the skew-t models generally improve over a Gaussian model. Thresholded models are not uniformly better in the simulation setting, which is expected because most simulated data are generated directly from the full skew-t model rather than from a censored tail model. The simulation therefore confirms that the skew-t framework provides flexibility for asymmetric and heavy-tailed data while remaining competitive with Gaussian methods when the data are light-tailed and symmetric.

The MRTS evidence is qualitatively similar to this benchmark picture: in settings 4 and 5, augmenting the regression component with MRTS basis functions can improve selected upper-tail Brier or quantile scores, but the gains are not stable across model families or quantile levels, and computation time grows visibly with K_{MRTS}, most clearly at $K_{\text{MRTS}} = 20$.

Figures **??** and **??** make this explicit by comparing methods 2–5 (skew-t and symmetric-t with $K = 1$ or $K = 5$ knots) at high threshold quantiles ($0.90 \leq q \leq 0.995$), with each figure showing two side-by-side panels: the left panel is the no-MRTS baseline and the right panel adds an MRTS basis of rank $K_{\text{MRTS}} = 15$. All scores are normalized by the Gaussian baseline (method 1) computed without MRTS, so values below the dashed reference line at 1 indicate better exceedance prediction than the Gaussian baseline. Setting 4 (skew-t with $K = 1$ knot, $\lambda = 3$) is shown in Figure **??**, and setting 5 (skew-t with $K = 5$ knots, $\lambda = 3$) in Figure **??**.

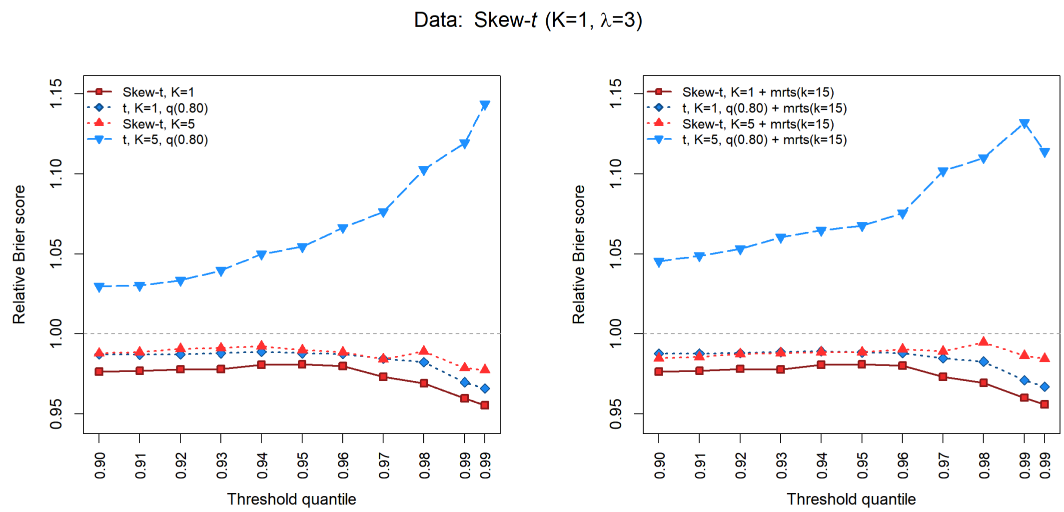


Figure 1: Setting 4 (data-generating process: skew-t with $K = 1$ knot, $\lambda = 3$). Relative Brier scores of methods 2–5 versus the Gaussian baseline at threshold quantiles $0.90 \leq q \leq 0.995$, with $K_{\text{MRTS}} = 15$. The left panel shows the no-MRTS baseline and the right panel shows the same methods with the MRTS basis added. The matching method (skew-t with $K = 1$, method 2) is best at every quantile, the over-specified five-knot threshold-t method (method 5) is worse than Gaussian, and the right panel is nearly indistinguishable from the left.

Two consistent patterns emerge. First, the data-generating knot count is the dominant axis: in setting 4 the one-knot skew-t method (method 2) is best at every quantile, and in setting 5 the five-knot skew-t method (method 4) is best at every quantile. Mis-specifying K on the under-side (fitting $K = 1$ to a $K = 5$ truth, as method 2 does in setting 5) gives only marginal gains over Gaussian, whereas mis-specifying K on the over-side (fitting $K = 5$ to a $K = 1$ truth, as method 4 does in setting 4) is competitive but not optimal. Second, the MRTS basis is essentially flat across the configurations studied here: in both settings the right (with-MRTS) panel reproduces the left (without-MRTS) panel almost exactly, and the same pattern was observed at the other ranks examined ($K_{\text{MRTS}} \in \{10, 20\}$, results not shown). The MRTS basis therefore does not change the within-setting ranking, and its added computational cost is not repaid by tail-score improvement under these data-generating processes.

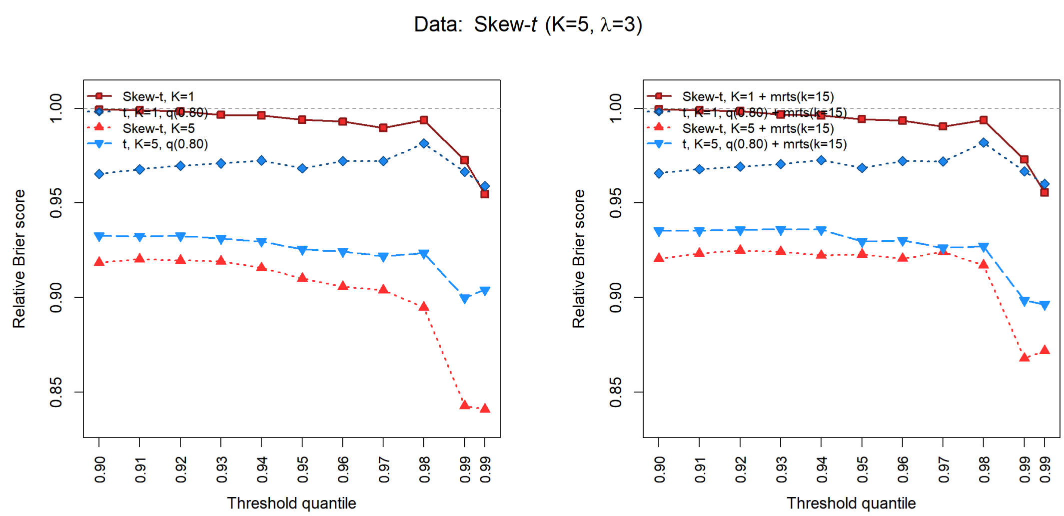


Figure 2: Setting 5 (data-generating process: skew-t with $K = 5$ knots, $\lambda = 3$). Relative Brier scores of methods 2–5 versus the Gaussian baseline; left panel is the no-MRTS baseline, right panel adds an MRTS basis of rank $K_{\text{MRTS}} = 15$. The matching five-knot skew-t method (method 4) is decisively the best across all quantiles, with relative Brier scores between roughly 0.84 and 0.92. All four methods improve on the Gaussian baseline, in contrast with setting 4 where the over-specified method 5 was worse than Gaussian. The right panel mirrors the left panel closely, indicating that the MRTS basis adds little beyond what the spatial skew-t structure already captures.

4.3 AR(2) on AR(2)-generated data: design

The AR(2) extension is evaluated in two experiments, each designed to give the temporal probe its most favorable test.

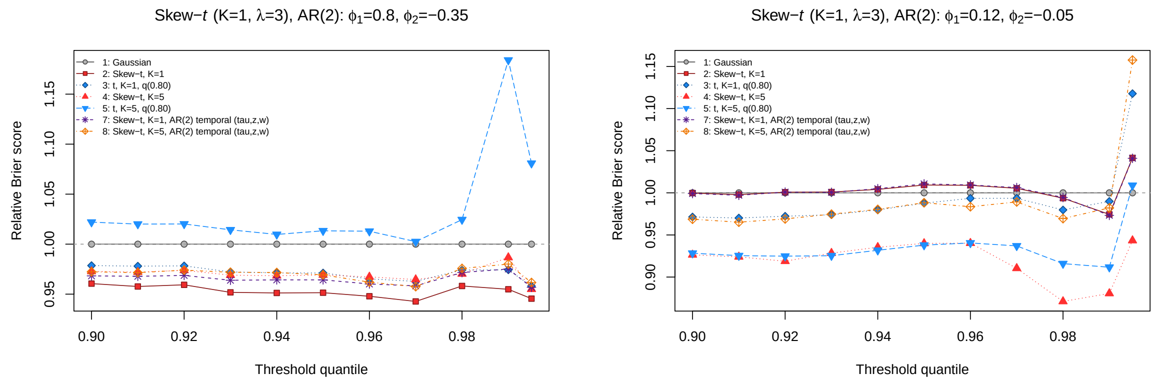
The first is a missing-at-location experiment with temporal dependence built into the latent τ, $\mathbf{w}$, and $\mathbf{z}$ processes. The setup reuses the Morris benchmark of Section **??**: $n_s = 144$ sites on $[0, 10] \times [0, 10]$ observed over $n_t = 50$ time points, with 100 sites used for training and 44 sites withheld for prediction, and 50 repetitions. Three temporal structures are compared on each repetition: no time series, AR(1), and AR(2). Each fitted model uses a skew-t marginal distribution with $K = 5$ knots and no thresholding, and the MCMC is run for $5\,000$ iterations with $2\,500$ burn-in iterations.

The second is a fixed-AR(2) experiment that embeds the temporal mechanism directly into the Morris benchmark of Section **??**, so the AR(2) analysis model can be tested against data with a known autoregressive structure on the same $n_s = 144$ sites and $n_t = 50$ time points. Two further data-generating settings are added. Both produce one-knot skew-t data ($K = 1$, $\lambda = 3$), but the latent τ, $\mathbf{w}$, and z processes evolve through a fixed AR(2) recursion (**??**)–(**??**) whose coefficients are held constant across time. Setting 9 uses a strongly dependent coefficient pair $(\phi_1, \phi_2) = (0.80, -0.35)$, and Setting 10 a weakly dependent pair $(\phi_1, \phi_2) = (0.12, -0.05)$; both pairs lie inside the stationarity triangle (**??**). For each setting, 50 data sets are simulated and split into 100 training and 44 validation sites. Each data set is fit by seven analysis methods: the five baseline models listed in Section **??**, together with two AR(2)-augmented skew-t models—one with $K = 1$ knot and one with $K = 5$ knots—whose time-varying coefficients follow the AR(2) prior (**??**)–(**??**). Because the data are themselves generated with AR(2) dependence, this comparison gives the temporal extension its most favorable test.

4.4 AR(2) on AR(2)-generated data: results

Figure **??** examines the AR(2) extension directly. All seven analysis methods are fitted to the two fixed-AR(2) data-generating settings, and for each method the Brier score is plotted relative to the Gaussian model across upper-tail threshold quantiles. A value below the dashed reference line at 1 means the method predicts exceedances better than the Gaussian baseline at that quantile, and a value above it means worse.

Under strong serial dependence (Setting 9, Figure **??**), every skew-t and symmetric-t method except the five-knot thresholded t-model improves on the Gaussian baseline, with rel-



(a) Setting 9: strong AR(2) dependence, $(\phi_1, \phi_2) = (0.80, -0.35)$.

(b) Setting 10: weak AR(2) dependence, $(\phi_1, \phi_2) = (0.12, -0.05)$.

Figure 3: Relative Brier score (versus the Gaussian model) by threshold quantile for the seven analysis methods fitted to the fixed-AR(2) data-generating settings. Values below the dashed reference line at 1 indicate better exceedance prediction than the Gaussian baseline; values above it indicate worse.

ative Brier scores between roughly 0.94 and 0.98. The one-knot skew-t model, which matches the $K = 1$ data-generating truth, is the best method, with a relative score near 0.95. Critically, adding the AR(2) temporal structure does not help: the one-knot AR(2) skew-t curve sits essentially on top of—and slightly above—the one-knot non-temporal skew-t curve, and the five-knot AR(2) model tracks its non-temporal counterpart. The five-knot thresholded t-model is the only method worse than Gaussian, and its relative score degrades sharply at the 0.99 quantile.

Under weak serial dependence (Setting 10, Figure **??**) the pattern shifts. The five-knot skew-t and five-knot thresholded t models are now clearly best, with relative Brier scores mostly between about 0.88 and 0.94, while the one-knot models—including the one-knot AR(2) model—are essentially indistinguishable from the Gaussian baseline. Once again the AR(2) and non-temporal curves nearly coincide at each knot count. Far-tail estimates are noisy: at the 0.995 quantile several methods rise above 1 because few observations exceed such extreme thresholds.

Two messages emerge. First, on data that are explicitly generated with AR(2) dependence—the most favorable possible setting for the temporal extension—fitting an AR(2) analysis model yields no measurable Brier-score gain over the corresponding non-temporal skew-t model at any quantile. Predictive accuracy is governed by the marginal and spatial flexibility of the model, namely the skew-t margin and the number of knots, rather than by the autoregressive order of the latent coefficients. Second, the preferred knot count depends on the dependence regime: strong temporal dependence favors the parsimonious one-knot model that matches the generating truth, whereas weak dependence favors the richer five-knot partition. This reinforces the conclusion that the AR(2) structure should be treated as a candidate option rather than a default improvement.

A complementary set of missing-at-location experiments places the AR(2), AR(1), and no-time-series specifications against a range of dependence regimes; the resulting Brier scores at the 0.90, 0.95, and 0.99 exceedance levels are collected in Table **??**. AR(2) improves mid-tail Brier scores in some scenarios – most visibly when one or more of the latent autoregressive coefficients is large – but AR(1) or the no-time-series baseline remains preferable in other scenarios or at specific tail levels, and the gains over AR(1) are typically smaller than the gap between any time-series model and the no-time-series baseline in runtime. Temporal flexibility therefore does not translate into uniform predictive dominance.

Scenario (missing-at-location)	BS90 (AR2/AR1/NoTS)	BS95 (AR2/AR1/NoTS)	BS99 (AR2/AR1/NoTS)	Time(s) (AR2/AR1/NoTS)
σ^2 AR(2) only	0.0129/0.0130/0.0128	0.0106/0.0106/0.0106	0.0032/0.0032/0.0032	54.78/54.53/51.41
All large ϕ	0.0344/0.0346/0.0352	0.0269/0.0267/0.0268	0.0071/0.0061/0.0069	138.47/135.74/109.64
All small ϕ	0.0267/0.0267/0.0266	0.0178/0.0178/0.0179	0.0059/0.0060/0.0060	232.74/228.68/187.20
Big ϕ_τ, others small	0.0388/0.0390/0.0389	0.0181/0.0182/0.0182	0.0045/0.0044/0.0043	135.17/134.59/102.93

Table 1: Selected AR(2) simulation results under missing-at-location settings; lower values indicate better predictive performance.

Table **??** juxtaposes the AR(2) and MRTS probes in a common decision summary, recording the empirical pattern and the practical implication for each.

Extension	Simulation setup	Main empirical pattern	Practical implication
AR(2) time-varying coefficients	Missing-at-location experiments with AR(2), AR(1), and no-time-series fitted to multiple dependence regimes	AR(2) can improve selected mid-tail Brier scores, but runtime advantages are often with AR(1) or no-time-series; no uniform winner across regimes	Use AR(2) as a candidate temporal model, not as a default replacement
MRTS covariates	Benchmark simulation settings 4–5 with additional multi-resolution thin-plate spline covariates ($K_{\text{MRTS}} = 10, 15, 20$ or 15)	Improvements appear at selected tail levels, but average differences are small and not stable across quantiles/model classes; runtime increases with larger K_{MRTS}	Use MRTS selectively when nonstationarity is plausible and tail-score gains justify added cost

Table 2: Integrated simulation summary for the two proposed extensions.

Overall, the simulation study indicates that both extensions enlarge the usable model class but that neither uniformly dominates the Morris baseline. The AR(2) prior provides temporal flexibility and the MRTS covariates enrich the spatial regression component, yet their empirical benefits depend on the data-generating mechanism and on the target exceedance level. Whether the same pattern holds on real environmental data—and, if so, what it implies for practitioners choosing among the candidate specifications—is the question taken up in the Data analysis section, where AR(2) and MRTS variants are treated as candidate options to be selected by held-out predictive scores rather than as default replacements for the AR(1) Morris model.

5 Real data analysis

The two case studies in this section apply the model comparison framework to real data with structurally different temporal regimes. The primary application is daily ground-level ozone (Section **??**), the dataset for which the Morris baseline was originally designed. As a controlled medium-size complement (Section **??**), we run the temporal probe on hourly MERRA-2 surface-temperature output, a regime that is smoother and less tail-dominated than ozone. Both case studies score posterior predictive distributions on held-out sites using the Brier and quantile scores defined in Section **??**.

5.1 EPA ground-level ozone

The real-data analysis applies the model comparison framework to daily maximum 8-hour ozone observations from July 2005. The response is the AQS monitor observation, and the main covariate is the corresponding CMAQ model output. Following the ozone analysis in Morris et al. (2017), the regression component uses $\mathbf{X}_t(\mathbf{s}) = [1, \mathrm{CMAQ}_t(\mathbf{s})]^\top$, so the model is interpreted as a statistical calibration of CMAQ using site-level observations. The full comparison data set contains 800 monitoring sites observed over 31 days. We use two-fold cross validation, with 400 sites used for training and 400 sites used for validation in each fold.

The purpose of this analysis is to test the simulation-based findings on a real dataset that the Morris framework was originally designed for. Concretely, the question is whether the two probe extensions—AR(2) on the temporal axis and MRTS on the spatial-mean axis—move the upper-tail Brier ranking when the comparison is run on the same EPA data that motivated the Morris specification. A preliminary MERRA-2 analysis with 100 sites and 100 time points already showed that AR(2), AR(1), and no-time-series models have very similar MAPE, Brier, and quantile scores, so the ozone comparison is set up not to declare a winner but to localize where, if anywhere, the enlarged candidate class improves held-out tail prediction.

5.1.1 Model comparisons

The candidate set contains approximately one hundred model settings. The baseline settings include Gaussian, skew-t , symmetric-t, and max-stable models, with different numbers of knots, censoring thresholds, and time-series choices. For the skew-t and symmetric-t models, the number of knots includes values such as $K = 1, 5, 6, 7, 8, 9, 10$, and 15. Censoring thresholds include no censoring, a moderate threshold such as $T = 50$, and a high threshold such as $T = 75$ ppb. The threshold 75 ppb is scientifically important because it corresponds to the air-quality exceedance level emphasized in the original ozone application.

The extended candidate set adds two families. The first replaces the AR(1) time-varying coefficient structure with the AR(2) structure developed in this thesis. The second adds MRTS covariates to selected Morris-type settings, so that nonstationary spatial structure can enter through multiple-resolution thin-plate spline basis functions. These extensions are compared against the original Morris-type specifications using the same cross-validation splits, so the ranking reflects predictive performance rather than differences in train-test construction.

For each fitted model, posterior predictive samples at the validation sites are used to compute Brier scores and quantile scores. The Brier scores are evaluated at upper-tail empirical thresholds $q(0.90)$, $q(0.95)$, $q(0.98)$, $q(0.99)$, and $q(0.995)$, and quantile scores are evaluated at the same probability levels. Scores are averaged over validation sites, days, and folds. Lower scores indicate better prediction.

Each MCMC chain is run for 30000 iterations with 25000 burn-in iterations. As in the simulation study, convergence of prediction summaries is more important here than direct interpretation of partition-specific latent variables, because multiple-partition models can suffer from label switching. The model-selection criterion is therefore based on held-out prediction scores, not on a visual comparison of latent partition labels.

5.1.2 Results

The real-data results support enlarging the candidate model class beyond the Morris baseline. Tables **??** and **??** list the top two models, by Brier score and by quantile score respectively, in the primary 400-train/400-validation comparison. AR(2) and MRTS variants appear repeatedly among the leading specifications: at the 0.90 Brier level the best model is an MRTS skew-t specification, only marginally ahead of the corresponding Morris specification; AR(2) models lead the Brier comparisons at 0.95 and 0.995 and the quantile-score comparisons at 0.98, 0.99, and 0.995. At intermediate exceedance levels, however, Morris skew-t or symmetric-t specifications remain strongest, so no single proposed extension dominates the entire tail.

Level	Rank 1 model	Score	Rank 2 model	Score
0.9000	**MRTS** skew-t, $K = 1$, $T = 0$, AR(1), 10 MRTS covariates	0.04467	Morris skew-t, $K = 1$, $T = 0$, AR(1)	0.04470
0.9500	**AR2** skew-t, $K = 7$, $T = 50$	0.02664	**AR2** skew-t, $K = 10$, $T = 50$	0.02672
0.9800	Morris skew-t, $K = 6$, $T = 50$, AR(1)	0.01257	Morris skew-t, $K = 5$, $T = 50$, AR(1)	0.01257
0.9900	Morris symmetric-t, $K = 9$, $T = 75$, AR(1)	0.00688	Morris skew-t, $K = 5$, $T = 50$	0.00690
0.9950	**AR2** symmetric-t, $K = 15$, $T = 75$	0.00384	Morris symmetric-t, $K = 9$, $T = 75$, AR(1)	0.00385

Table 3: Top two Brier-score models in the US-all ozone validation study. Values in parentheses are held-out scores from the comparison workbook; smaller values indicate better predictive performance.

Level	Rank 1 model	Score	Rank 2 model	Score
0.9000	**MRTS** skew-t, $K = 1$, $T = 0$, AR(1), 5 MRTS covariates	5.033	Morris skew-t, $K = 5$, $T = 0$	5.033
0.9500	Morris skew-t, $K = 7$, $T = 50$, AR(1)	2.988	**AR2** skew-t, $K = 10$, $T = 0$	2.988
0.9800	**AR2** skew-t, $K = 10$, $T = 0$	1.423	Morris skew-t, $K = 9$, $T = 0$	1.424
0.9900	**AR2** skew-t, $K = 6$, $T = 50$	0.797	Morris skew-t, $K = 9$, $T = 50$, AR(1)	0.798
0.9950	**AR2** skew-t, $K = 6$, $T = 50$	0.441	**AR2** skew-t, $K = 8$, $T = 50$	0.442

Table 4: Top two quantile-score models in the US-all ozone validation study. Values in parentheses are held-out scores from the comparison workbook; smaller values indicate better predictive performance.

The ranking does not concentrate on a single universally optimal specification. Among the Brier-score leaders, the strongest settings typically combine CMAQ, a time-series component, a moderate-to-high censoring threshold, and a small-to-moderate number of knots. Among the quantile-score leaders, AR(2) specifications appear frequently, particularly at $q(0.95)$ and above. MRTS specifications are more selective: they improve certain tail exceedance scores but are not uniformly preferred across all score types or quantile levels.

To gauge how stable these rankings are to the train/validation split, we repeat the comparison at two smaller training sizes. In the primary 400/400 split the empirical Brier thresholds at 0.90, 0.95, 0.98, 0.99, and 0.995 are 73.25, 79.75, 88.50, 94.75, and 100.52 ppb, and the top

Level	Rank 1	Rank 2	Rank 3	Rank 4
0.9000	204 (MRTS, 0.041)	55 (Morris's, 0.041)	205 (MRTS, 0.041)	70 (Morris's, 0.041)
0.9500	70 (Morris's, 0.020)	41 (Morris's, 0.020)	12 (Morris's, 0.020)	39 (Morris's, 0.020)
0.9800	204 (MRTS, 0.008)	117 (AR2, 0.008)	67 (Morris's, 0.008)	111 (AR2, 0.008)
0.9900	117 (AR2, 0.003)	15 (Morris's, 0.003)	70 (Morris's, 0.003)	61 (Morris's, 0.003)
0.9950	15 (Morris's, 0.001)	70 (Morris's, 0.001)	61 (Morris's, 0.001)	117 (AR2, 0.001)

Table 5: Validation selection result using 300 sites to predict 100 sites

Level	Rank 1	Rank 2	Rank 3	Rank 4
0.9000	54 (Morris's, 0.050)	105 (AR2, 0.050)	35 (Morris's, 0.050)	215 (MRTS, 0.050)
0.9500	117 (AR2, 0.030)	116 (AR2, 0.030)	114 (AR2, 0.030)	70 (Morris's, 0.030)
0.9800	215 (MRTS, 0.015)	205 (MRTS, 0.015)	111 (AR2, 0.015)	58 (Morris's, 0.015)
0.9900	58 (Morris's, 0.008)	112 (AR2, 0.008)	216 (AR2, 0.008)	67 (Morris's, 0.008)
0.9950	29 (Morris's, 0.004)	35 (Morris's, 0.004)	60 (Morris's, 0.004)	116 (AR2, 0.004)

Table 6: Validation selection result using 200 sites to predict 200 sites

settings at these levels are numbered 204, 111, 58, 68, and 124 in the candidate list. When the split is changed to 300 training and 100 validation sites, the corresponding leaders become 204, 70, 204, 117, and 15; with 200 training and 200 validation sites they become 54, 117, 215, 58, and 29. The same broad model families (Morris skew-t , AR(2), and MRTS variants) remain competitive across all three splits, but the exact best-ranked setting shifts with the split and the exceedance level. Such instability is expected: at the 0.99 and 0.995 levels only a handful of validation observations exceed the threshold, so the ranking is sensitive to which sites happen to fall in the held-out set. The real-data analysis therefore should not be read as identifying a single universally optimal specification, but as evidence that AR(2) and MRTS variants are valuable additions to the candidate pool when the inferential target is held-out tail prediction.

Overall, the real-data analysis reinforces the simulation pattern: the two probes do not displace the Morris baseline but redraw the leaderboard on the upper tail. MRTS, the spatial-mean probe, appears among the top settings at several Brier and quantile levels and is the more empirically convincing of the two extensions. AR(2), the temporal probe, appears among the leading models for several upper-tail quantile scores but does not deliver a robust Brier-score improvement over AR(1)—an empirical pattern that the Discussion explains as a structural consequence of the Morris kernel rather than an accident of this particular dataset. The practical recommendation is therefore to keep the Morris baseline as the default specification and to admit AR(2) and MRTS variants into the candidate pool, with the selection among them made by held-out Brier and quantile scores on the application of interest. The next subsection asks whether the AR(2)-versus-AR(1) verdict survives outside the ozone setting, on a controlled hourly-temperature benchmark whose dynamics are smoother and less tail-dominated.

5.2 MERRA-2 surface temperature

To complement the ozone case study with a controlled medium-size benchmark on data that is structurally different from EPA ozone, we run the temporal probe on hourly surface-temperature output from the MERRA-2 reanalysis. The question is whether the AR(2) extension yields measurable Brier-score gains in a regime that is smoother and less tail-dominated than ozone—in other words, whether the temporal axis of the Morris hierarchy carries more leverage when the underlying field is itself temporally smoother. The data comprise $n_s = 100$ spatial locations observed at $n_t = 100$ consecutive hourly time points, with longitude and latitude retained for each site.

5.2.1 Data preprocessing and model setup

We randomly split the sites into training and validation subsets using seed 123, allocating 70% of the sites to model fitting and the remaining 30% to held-out evaluation. The regression design uses an intercept and a first-order spatial trend,

$$\mathbf{X}_t(\mathbf{s}) = [1,\ s_1,\ s_2]^\top, \tag{30}$$

where s_1 and s_2 are the longitude and latitude of site $\mathbf{s}$. Because the covariates are not standardized in this experiment, we set `nu.upper = 10` so that the smoothness parameter ν of the Matérn covariance cannot drift into the numerically unstable large-ν regime during MCMC.

To isolate the effect of temporal structure, the three temporal specifications – no time series, AR(1), and AR(2) – share the same core configuration: a skew-t margin, a Matérn covariance, $K = 5$ knots, and no censoring threshold. Each chain is run for $20\,000$ iterations with the first $10\,000$ discarded as burn-in, yielding $10\,000$ retained posterior draws per model.

5.2.2 Results

Table **??** reports quantile-score (QS) and Brier-score (BS) comparisons across the three temporal specifications. The differences are small in absolute terms but qualitatively informative:

the no-time-series model performs marginally better on the QS metrics and attains the lowest BS at the most extreme exceedance level (0.99), whereas AR(2) attains the best BS at the 0.90 and 0.95 levels. None of the three specifications therefore dominates uniformly; the preferred model depends on whether one targets bulk predictive accuracy, mid-tail exceedance probabilities, or the far tail.

The remaining summary metrics reinforce this reading. MAPE is slightly lower for AR(2) (0.0106), with AR(1) close behind, while the no-time-series model is somewhat worse on this bulk-accuracy measure. The 95% predictive-interval coverage is nearly identical across the three models (0.912–0.916), suggesting that temporal dependence affects predictive sharpness more than calibration. These modest gains come at a non-trivial computational cost: total fitting time rises from 777.3 seconds for the no-time-series model to 1041.7 seconds for AR(1) and 1075.8 seconds for AR(2). Taken together, the MERRA-2 results suggest that AR(2) is a defensible option when slightly better mid-tail behavior is desired, but the improvement is incremental rather than decisive—consistent with the picture obtained from the simulation study and the EPA ozone analysis. That the same incremental verdict appears across three datasets of different size, signal-to-noise, and temporal smoothness motivates the structural explanation given in the Discussion section that follows.

Model	QS90	QS95	QS99	BS90	BS95	BS99
No Temporal	**4.714**	**2.522**	**0.570**	0.041	0.018	**0.008**
AR(1)	4.716	2.522	0.570	0.040	0.016	0.009
AR(2)	4.717	2.522	0.571	**0.040**	**0.016**	0.009

Table 7: Comparison of temporal specifications in the MERRA-2 experiment. Smaller values indicate better predictive performance.

6 Discussion

The simulation study, the EPA ozone analysis, and the MERRA-2 benchmark of the preceding sections produced the same qualitative picture: the AR(2) extension is a defensible alternative to the AR(1) baseline but does not systematically improve the upper-tail Brier score, with relative gains at the level of Monte Carlo noise. Across three datasets that differ in sample size, signal-to-noise, and temporal smoothness, the empirical pattern is robust enough to demand a structural explanation rather than a dataset-specific one. This section records that explanation. The argument turns on three ingredients: (i) the unit-variance standardization built into (**??**)–

(**??**) makes the time-t marginal of every latent invariant to (ϕ_1, ϕ_2); (ii) the predictive density at a held-out site–day depends on the latent field only through its time-t slice; and (iii) the variance of the predictive level decomposes into a (ϕ_1, ϕ_2)-invariant floor and a partial AR-reducible term, the latter further attenuated by the censoring and partition mechanisms.

Notation. Throughout this section we write $\phi := (\phi_1^{(\cdot)}, \phi_2^{(\cdot)})$ generically for the AR(2) coefficients of any latent series and $\boldsymbol{\theta}$ for the remaining fixed-parameter vector ($\boldsymbol{\beta}$, λ, φ, ν, γ, and the partition hyperparameters a, b). Let $\mathbf{Y}^{\text{tr}}$ denote the censored training observations $\{\widetilde{Y}_t(\mathbf{s})\}$ at training sites $\mathbf{s}$ and $t = 1, \ldots, n_t$. Let $g(\mathbf{s}^*, t) \in \{1, \ldots, K\}$ be the partition index of the held-out site $\mathbf{s}^*$ on day t, defined by $\mathbf{s}^* \in P_{t, g(\mathbf{s}^*, t)}$, and write $g := g(\mathbf{s}^*, t)$ for short. Stack the day-t knot-level latents into

$$ {}_{t} \;:=\; \big(z_{t1}, \ldots, z_{tK},\; \sigma_{t1}^2, \ldots, \sigma_{tK}^2,\; \mathbf{w}_{t1}, \ldots, \mathbf{w}_{tK}\big). \tag{31} $$

For a threshold c the Brier score at the held-out cell $(\mathbf{s}^*, t)$ is

$$ \mathrm{BS}(\mathbf{s}^*, t; c) \;:=\; E\Big[\big(\mathbf{1}\{Y_t(\mathbf{s}^*) > c\} - \hat{p}(\mathbf{s}^*, t; c)\big)^2\Big], \qquad \hat{p}(\mathbf{s}^*, t; c) \;:=\; \Pr\big(Y_t(\mathbf{s}^*) > c \mid \mathbf{Y}^{\text{tr}}\big). \tag{32} $$

Marginal invariance to ϕ

The standardization that fixes $\gamma_0 = \mathrm{Var}(Y_t) = 1$ in (**??**)–(**??**), together with the bivariate stationary initialization (**??**), has a consequence that is decisive for the held-out Brier score in (**??**).

Proposition 1 (Marginal invariance to ϕ)**.** *Let $\{Y_t\}_{t=1}^{n_t}$ be generated by the standardized AR(2) recursion (**??**) with innovation variance (**??**) and stationary initial pair (**??**), under the stability condition (**??**). Then $Y_t \sim N(0, 1)$ for every $t \in \{1, \ldots, n_t\}$, regardless of (ϕ_1, ϕ_2).*

Sketch. By (**??**), $Y_1, Y_2 \sim N(0, 1)$. Suppose inductively that $\mathrm{Var}(Y_{t-1}) = \mathrm{Var}(Y_{t-2}) = 1$ and $\mathrm{Cov}(Y_{t-1}, Y_{t-2}) = \gamma_1$. Substituting (**??**) into $\mathrm{Var}(Y_t) = \phi_1^2 + \phi_2^2 + 2\phi_1\phi_2\gamma_1 + \sigma_\epsilon^2$ and using (**??**)–(**??**) gives $\mathrm{Var}(Y_t) = 1$. Means are zero by linearity, and joint normality is preserved by the linear recursion. □

Because the inverse PIT layer (**??**)–(**??**) that maps each $Y_t = w_{tki}^*$, z_{tk}^*, or σ_{tk}^{2*} back to the working scale is a fixed monotone function of Y_t alone, Proposition **??** carries over verbatim:

Corollary 1 (ϕ-free latent marginal at each t)**.** *Under (**??**)–(**??**) for the transformed latents $z_{tk}^*, \sigma_{tk}^{2*}, \mathbf{w}_{tk}^*$, the prior marginal of ${}_{t}$ in (**??**) does not depend on ϕ for any t. In particular, the*

AR(1) baseline of Morris et al. (2017) and the AR(2) extension of Section **??** *share the same prior marginal at every fixed time point.*

The AR order is invisible to the one-point marginal; it is identifiable only from *joint* laws across time.

The Brier score sees only the time-t marginal

The kernel (**??**) at a single space–time cell $(\mathbf{s}^*, t)$ depends on the latents only through ${}_t$, so the predictive exceedance probability factorizes as

$$\hat{p}(\mathbf{s}^*, t; c) \;=\; \iint \Pr\!\big(Y_t(\mathbf{s}^*) > c \,\big|\, \boldsymbol{\theta}, {}_t\,\big)\, \pi\big({}_t \,\big|\, \boldsymbol{\theta}, \mathbf{Y}^{\mathrm{tr}}\big)\, \pi\big(\boldsymbol{\theta} \,\big|\, \mathbf{Y}^{\mathrm{tr}}\big)\, \mathrm{d}_t\, \mathrm{d}\boldsymbol{\theta}. \tag{33}$$

Combining (**??**) with Corollary **??** gives the following structural fact.

Proposition 2 (ϕ enters Brier only through the posterior)**.** *For the kernel (***??***) and the latent prior of Section* **??***, ϕ can influence the held-out Brier score (***??***) only through two routes: the posterior $\pi(\boldsymbol{\theta} \mid \mathbf{Y}^{\mathrm{tr}})$ over the hyperparameters and the conditional $\pi({}_t \mid \boldsymbol{\theta}, \mathbf{Y}^{\mathrm{tr}})$ obtained by temporal pooling. The prior marginal of ${}_t$ at the test day is ϕ-blind.*

The first route (posterior on $\boldsymbol{\theta}$) adds estimation variance proportional to the size of the augmented parameter space; the second route (temporal pooling of ${}_t$) is the only mechanism by which a richer AR order can *tighten* the predictive. The AR(2) prior introduces two extra coefficients per latent series and thereby pays a cost on the first route; the question is whether the second route delivers a benefit large enough to overcome it.

Reducible / irreducible decomposition and the variance floor

Write $p(c) := \Pr(Y_t(\mathbf{s}^*) > c)$ for the true exceedance probability. Standard manipulation of (**??**) gives the bias–variance decomposition

$$E\big[\mathrm{BS}(\mathbf{s}^*, t; c)\big] \;=\; \underbrace{p(c)\,(1 - p(c))}_{\text{irreducible}} \;+\; \underbrace{\big(E[\hat{p}(\mathbf{s}^*, t; c)] - p(c)\big)^2 \;+\; \mathrm{Var}\big(\hat{p}(\mathbf{s}^*, t; c)\big)}_{\text{reducible by the model}}. \tag{34}$$

The irreducible term is a property of the data-generating process and is ϕ-invariant; AR(2) can act only on the reducible term. When each knot is informed by a sizeable group of training sites the likelihood swamps the prior, both models are unbiased to leading order, and the contest reduces to $\mathrm{Var}(\hat{p})$. A delta-method expansion of $\hat{p}$ around the predictive mean of $Y_t(\mathbf{s}^*)$ gives

$$\mathrm{Var}\big(\hat{p}(\mathbf{s}^*, t; c)\big) \;\approx\; \big[f_{Y_t(\mathbf{s}^*)|\mathbf{Y}^{\mathrm{tr}}}(c)\big]^2\, \mathrm{Var}\big(Y_t(\mathbf{s}^*) \,\big|\, \mathbf{Y}^{\mathrm{tr}}\big), \tag{35}$$

where $f_{Y_t(\mathbf{s}^*)|\mathbf{Y}^{\mathrm{tr}}}$ is the predictive density at the held-out cell. To localize the ϕ-dependence in (**??**) we decompose the predictive variance by conditioning. Define the spatial kriging-residual variance at the held-out site

$$\rho^*_{\mathbf{s}^*,t} \;:=\; \mathrm{Var}\big(v_t(\mathbf{s}^*) \,\big|\, \boldsymbol{\theta},\, \mathbf{Y}^{\mathrm{tr}}\big), \tag{36}$$

which is determined entirely by the Matérn hyperparameters (φ, ν, γ) and the spatial configuration of training sites; in particular it does not depend on ϕ. Conditional on $(\boldsymbol{\theta}_{,t}\,, \mathbf{Y}^{\mathrm{tr}})$ the kernel (**??**) gives

$$Y_t(\mathbf{s}^*) \,\big|\, \boldsymbol{\theta}_{,t}\,, \mathbf{Y}^{\mathrm{tr}} \;\sim\; N\Big(\mathbf{X}_t(\mathbf{s}^*)^\top \boldsymbol{\beta} + \lambda\, \sigma_{tg}\, |z_{tg}| + \sigma_{tg}\, \hat{v}_t(\mathbf{s}^*),\; \sigma^2_{tg}\, \rho^*_{\mathbf{s}^*,t}\Big), \tag{37}$$

where $\hat{v}_t(\mathbf{s}^*) := E[v_t(\mathbf{s}^*) \mid \boldsymbol{\theta}, \mathbf{Y}^{\mathrm{tr}}]$ is the kriging mean. Applying the law of total variance with respect to the posterior on $(\boldsymbol{\theta}_{,t}\,)$ then yields

$$\mathrm{Var}\big(Y_t(\mathbf{s}^*) \,\big|\, \mathbf{Y}^{\mathrm{tr}}\big) \;=\; \underbrace{E\big[\sigma^2_{tg}\, \rho^*_{\mathbf{s}^*,t} \,\big|\, \mathbf{Y}^{\mathrm{tr}}\big]}_{\text{Channel D: spatial-noise floor}} \;+\; \underbrace{\mathrm{Var}\big(\mathbf{X}_t(\mathbf{s}^*)^\top \boldsymbol{\beta} + \lambda\, \sigma_{tg}\, |z_{tg}| + \sigma_{tg}\, \hat{v}_t(\mathbf{s}^*) \,\big|\, \mathbf{Y}^{\mathrm{tr}}\big)}_{\text{Channels A–B: } z\text{- and } \sigma\text{-pooling}}. \tag{38}$$

Proposition 3 (ϕ-invariant noise floor)**.** *For each fixed $\boldsymbol{\theta}$ the kriging-residual variance $\rho^*_{\mathbf{s}^*,t}$ in (**??**) is a function of the spatial hyperparameters and the spatial configuration alone, with no dependence on ϕ. Consequently, the floor term*

$$\mathrm{Var}\big(Y_t(\mathbf{s}^*) \,\big|\, \mathbf{Y}^{\mathrm{tr}}\big) \;\geq\; E\big[\sigma^2_{tg}\, \rho^*_{\mathbf{s}^*,t} \,\big|\, \mathbf{Y}^{\mathrm{tr}}\big] \tag{39}$$

can be moved by ϕ only through the AR(2) effect on the posterior mean $E[\sigma^2_{tg} \mid \mathbf{Y}^{\mathrm{tr}}]$.

Sketch. $v_t(\mathbf{s}^*) \mid \boldsymbol{\theta}$ is centered Gaussian, independent of $_t$, and i.i.d. across t. Its kriging-residual variance (**??**) depends only on the spatial covariance entries and the training configuration; the AR(2) prior on $\mathbf{w}^*_{tk}, z^*_{tk}, \sigma^{2*}_{tk}$ enters (**??**) only through $_t$. The lower bound (**??**) then follows by dropping the non-negative second summand in (**??**). □

The implication of Proposition **??** is that, even with a perfectly known ϕ that drives

$\mathrm{Var}(z_{tg} \mid \mathbf{Y}^{\mathrm{tr}})$ and $\mathrm{Var}(\sigma^2_{tg} \mid \mathbf{Y}^{\mathrm{tr}})$ as low as possible, the predictive variance entering (**??**) still carries the spatial-noise term in (**??**), which is ϕ-blind apart from the modulation of $E[\sigma^2_{tg} \mid \mathbf{Y}^{\mathrm{tr}}]$. In regimes where this floor is the dominant contributor to (**??**)—which is the case for both the simulated and the MERRA-2 settings, where the per-knot site count is large and the spatial correlation at $\mathbf{s}^*$ is moderate—the leverage of AR order on the Brier score is structurally small.

Why the gain in Channels A–B is also small

Within Channel A, the AR(2) prior tightens $\pi(z^*_{tg} \mid \mathbf{Y}^{\mathrm{tr}})$ by borrowing strength from $z^*_{(t\pm1)g}$ and $z^*_{(t\pm2)g}$. A Gaussian surrogate calculation on the standardized latent gives the following bound. Suppose the day-t likelihood contributes effective precision q to z^*_{tg} (so the i.i.d./AR(1)-equivalent posterior variance at t is $(1+q)^{-1}$), and let

$$\kappa(\phi) \;:=\; \frac{1+\phi_1^2+\phi_2^2}{\sigma_\epsilon^2} \tag{40}$$

be the AR(2) full-conditional precision read off (**??**) and (**??**). Then the AR(2) marginal posterior variance $v_t(\phi)$ satisfies

$$0 \;\le\; v_t^{\mathrm{iid}} - v_t(\phi) \;\le\; \frac{\kappa(\phi)-1}{(1+q)\,(\kappa(\phi)+q)} \;=\; \mathcal{O}\big(q^{-2}\big), \tag{41}$$

where $v_t^{\mathrm{iid}} := (1+q)^{-1}$ is the i.i.d. posterior variance and $\kappa(\phi) \ge 1$ with equality iff $\phi = 0$. In the data-rich regime ($q \gg 1$) the AR(2)-induced reduction in $\mathrm{Var}(z^*_{tg} \mid \mathbf{Y}^{\mathrm{tr}})$ scales as $1/q^2$, while the i.i.d./AR(1) variance scales as $1/q$. The same bound applies to σ^{2*}_{tg} along Channel B with the appropriate effective precision.

The censoring layer that retains only $\widetilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$ further attenuates the effective precision feeding σ^{2*}_{tg} and z^*_{tg}, because below-threshold observations contribute information only through the imputed values; this shrinks the per-day Fisher information for the scale latent in particular, and the bound (**??**) is then tight against an even smaller q.

The cost of ϕ via the posterior on $\boldsymbol{\theta}$

A complementary observation, made quantitative through (**??**), is that adding $\left(\phi_1^{(z)}, \phi_2^{(z)}, \phi_1^{(\sigma)}, \phi_2^{(\sigma)}, \phi_1^{(w)}, \phi_2^{(w)}\right)$ to the parameter vector inflates the posterior $\pi(\boldsymbol{\theta} \mid \mathbf{Y}^{\text{tr}})$. By the law of total variance,

$$\operatorname{Var}\!\big(\hat{p}(\mathbf{s}^*, t; c)\big) \;\approx\; E_\phi\!\big[\operatorname{Var}\!\big(\hat{p} \,|\, \phi\big)\big] \;+\; \operatorname{Var}_\phi\!\big(E[\hat{p} \,|\, \phi]\big). \tag{42}$$

The AR(2) prior shrinks the first summand through (**??**) (Channels A–B) but introduces the second summand, which is identically zero only when ϕ has a point posterior. When ϕ is weakly identified from the data—as is typical for $\phi^{(\sigma)}$ and $\phi^{(w)}$, where the per-day information about a scale parameter or a knot location read through an arg min is small—the posterior on ϕ stays diffuse and the second summand can offset the first. This is the route by which a model with more flexibility can produce a slightly *worse* Brier score than a more constrained one, and it is consistent with the empirical pattern in Table **??** and Table **??**, where the AR(2) variant beats the AR(1) baseline on some quantiles and loses on others without dominating overall.

What the analysis does and does not say

Three points are worth emphasizing. First, Propositions **??**–**??** are statements about the held-out *Brier* score at a single threshold; they do not preclude AR(2) gains under scoring rules that probe joint distributions across time (e.g., multi-day energy scores or rules sensitive to volatility clustering), nor do they apply to forecasting tasks in which one or more held-out days are extrapolated beyond the training window. Second, the variance bound (**??**) is non-asymptotic but expressed in the standardized-latent surrogate; converting it to a calibrated bound on the original-scale predictive requires propagating the inverse PIT (**??**)–(**??**), which only *attenuates* the bound further because the half-normal and inverse-gamma transforms compress the latent variability. Third, the noise floor (**??**) can be eroded by reducing the kriging-residual variance $\rho^*_{\mathbf{s}^*,t}$—through a more flexible spatial trend or the MRTS enrichment of the regression component—rather than by enriching the temporal prior. This is consistent with the simulation finding that MRTS variants out-perform AR(2) on bulk quantile scores: the leverage on held-out Brier lives in the spatial regression component, not in the temporal prior.

In summary, the standardization that is required to keep the marginal skew-t structure intact under the AR(2) extension is precisely what dissolves the AR-order signal at the held-out test day, and the residual route through posterior pooling of latents delivers an $\mathcal{O}(q^{-2})$

improvement on Channels A and B that is bounded below by the ϕ-invariant spatial-noise floor in (**??**). The empirical pattern observed in the simulation study, the EPA ozone analysis, and the MERRA-2 benchmark is therefore not an artifact of the MCMC implementation; it is the predicted behavior of the predictive Brier integral under the Morris kernel (**??**) combined with the unit-variance AR(2) prior (**??**)–(**??**). Because the argument depends only on (a) a unit-variance standardization of the temporal latents and (b) PIT-preserved marginals, it applies to any hierarchical extreme-value model that shares these two design choices, and it identifies the spatial regression component, rather than the temporal prior, as the structural place where future upper-tail-predictive improvement of this model class should be sought.

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